

Advanced Demand Forecasting Tool – Complete Documentation

1 . **Project Overview**

This project is an industry-style **Demand Forecasting Tool** built using Python and Streamlit.

It predicts future sales for each SKU using:

- Statistical Models
- Machine Learning
- Hybrid Forecasting

The system is designed for **Retail / E-commerce / Supply Chain use cases**.

2 . **Objective**

To forecast future demand at SKU level using:

- Historical sales patterns
- Seasonal behaviour
- Trend learning
- Residual correction

The tool helps answer:

➡ What will sell next? ➡ When demand will rise or fall? ➡ How accurate is our forecast?

3 . **Forecasting Approach Used**

This system uses a **Hybrid Forecasting Framework**.

Instead of depending on one model, we combine:

Model	Role
SARIMA	Stable baseline trend
Prophet	Seasonality + trend

Model	Role
XGBoost	Residual correction
Ensemble	Balanced forecast

4 . Data Requirements

The system accepts CSV data with flexible naming.

Minimum Required:

Column	Description
Date	Transaction date
SKU	Product ID
Sales / Qty	Units sold

Optional columns can be added later:

- Price
- Promotions
- Holiday flags

5 . Data Preprocessing

Steps performed automatically:

5 . 1 Column Standardization

User CSV column names are auto-detected.

Example:

Input	Converted To
order_date	Date
item_id	SKU
qty	Sales

5 . 2 Date Handling

Supports:

- DD-MM-YYYY
- YYYY-MM-DD
- Mixed formats

Handled using:

```
pd.to_datetime(..., dayfirst=True)
```

5 . 3 SKU Filtering

Forecasting is performed **SKU-wise**.

6 . Feature Engineering (for ML Model)

Generated features:

Feature		Purpose
Month		Seasonal signal
Day of Week		Weekly behaviour
Winter Flag		Demand seasonality
Lag	7	Short-term memory
Lag	3 0	Monthly memory
Rolling Mean	7	Trend smoothing
Rolling Mean	3 0	Demand baseline
Deviation from Avg		Mean reversion

7 . Models Used

7 . 1 SARIMA (Statistical Model)

Used for:

✓ Stable trend capture ✓ Seasonality ✓ Noise reduction

Works well when: - Patterns are smooth

Limitations: - Slow to react to sudden demand shifts

7 . 2 Prophet (Trend + Seasonality)

Developed by Meta.

Captures:

✓ Yearly patterns ✓ Seasonality ✓ Trend changes

Used as: ➡ Base demand signal

7 . 3 XGBoost (Machine Learning)

Used for:

✓ Capturing sudden changes ✓ Learning residual demand

Important:

XGBoost does NOT predict demand directly.

Instead it learns:

Residual = Actual Demand - Prophet Trend

This prevents: ✗ Artificial growth ✗ Spike continuation

8 . 🔄 Hybrid Forecasting Logic

Instead of predicting demand directly:

Step 1 : Prophet predicts trend

Step 2 : XGBoost predicts deviations

Final Forecast:

Forecast = Prophet Trend + XGBoost Residual

This stabilizes forecasts.

9 . Ensemble Model

Final ensemble combines:

- SARIMA
- Prophet
- Hybrid XGBoost

To reduce risk of model bias.

10 . Accuracy Measurement

We use:

MAPE (Mean Absolute Percentage Error)

Formula:

$$\text{MAPE} = \text{Average}(|\text{Actual} - \text{Forecast}| / \text{Actual}) \times 100$$

Interpretation:

MAPE	Accuracy
< 10 %	Excellent
10 – 20 %	Good
20 – 30 %	Acceptable
> 30 %	Needs Improvement

1 1 . Visualization

The tool displays:

- Historical Sales
- Forecasted Demand

Using Plotly for business clarity.

1 2 . Forecast Workflow

- 1 . Upload data
 - 2 . Select SKU
 - 3 . Select Model
 - 4 . Generate Forecast
 - 5 . View Accuracy
-

1 3 . Real-World Use Cases

Applicable for:

- Inventory planning
 - Seasonal product forecasting
 - Fashion retail
 - E-commerce
 - Supply chain optimization
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1 4 . Business Value

Helps in:

✓ Reducing stockouts ✓ Avoiding overstock ✓ Planning procurement ✓ Improving sales planning

1 5 . Future Improvements

Possible upgrades:

- Price elasticity
- Promotion impact

- Holiday forecasting
 - Auto model selection
 - Confidence intervals
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1 6 . Conclusion

This system is not a basic forecast model.

It is a hybrid demand prediction engine combining:

- Statistical modeling
- Machine learning
- Trend decomposition

Designed to simulate real retail analytics systems.
